

Optimizing Semantic Retrieval for Bengali: A Comparative Analysis of Monolingual and Multilingual Embeddings with Matryoshka Representation Learning

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Effective semantic retrieval remains the primary bottleneck for Bengali Retrieval-Augmented Generation (RAG) systems. While general-purpose multilingual models exist, they often lack the specific semantic alignment required for high-precision tasks. This paper conducts a comparative study of three distinct embedding architectures—monolingual (shihab17), distilled multilingual (distiluse), and paraphrase-focused (mpnet)—to identify the optimal foundation for Bengali retrieval. We fine-tune these models on the BanglaRQA dataset using a composite objective of Multiple Negatives Ranking Loss and Matryoshka Representation Learning (MRL). The results demonstrate that architectural choice is decisive: the fine-tuned mpnet model achieves a superior NDCG@10 of **0.8114**, statistically outperforming the monolingual baseline ($p < 0.001$). Beyond raw performance, we address the deployment constraints typical of low-resource environments. Our efficiency analysis reveals that high-dimensional vectors are largely redundant for this task; the MRL-trained MPNet model retains **96%** of its retrieval effectiveness at just **128 dimensions**, offering an 83% reduction in storage costs without compromising accuracy.

CCS Concepts: • **Computing methodologies** → **Natural language processing**; *Neural networks*; • **Information systems** → **Retrieval models and ranking**.

Additional Key Words and Phrases: Bengali Language, Sentence Embeddings, Fine-Tuning, Information Retrieval, Matryoshka Representation Learning, RAG

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1 Introduction

Large Language Models (LLMs) have demonstrated transformative capabilities across a spectrum of natural language tasks. However, their practical deployment is often hampered by two critical weaknesses: a tendency to “hallucinate” facts not present in their training data and an inability to access timely or proprietary information. Retrieval-Augmented Generation (RAG) has emerged as the dominant architectural paradigm to address these shortcomings [1, 2]. By dynamically retrieving relevant context from an external knowledge base before generation, RAG systems ground LLM outputs in factual reality, significantly mitigating hallucinations and enhancing their utility.

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The efficacy of any RAG system, however, hinges critically on its retrieval component, which is typically powered by dense vector embeddings. While this technology is mature for high-resource languages like English, it remains a significant bottleneck for the vast majority of the world’s languages. This challenge is particularly acute for Bengali, a language spoken by over 300 million people. Despite its large speaker base, Bengali is considered a “low-resource” language in the context of deep learning, suffering from a scarcity of curated datasets, commercially incentivized models, and research focus compared to English. Consequently, the choice of an embedding model is often driven by availability rather than empirical performance.

While multilingual models like XLM-RoBERTa [7] provide a strong foundation, they are often generalists that lack the specific semantic alignment required for question-answering tasks. Conversely, monolingual models may capture linguistic nuances but lack the architectural robustness of larger multilingual checkpoints. This dichotomy presents a research gap: identifying which architecture yields the highest fidelity for Bengali semantic retrieval when subjected to specialized fine-tuning.

This paper addresses this gap through a comprehensive study. We select three diverse architectures: a dedicated Bengali transformer, a lightweight distilled multilingual model, and a robust MPNet-based model. We subject all three to an identical fine-tuning regimen on the BanglaRQA dataset [8], employing Matryoshka Representation Learning (MRL) [9] to enforce efficiency. Our contributions are threefold. First, we provide a comparative benchmark of these architectures, demonstrating that while all benefit from fine-tuning, the architectural priors of the mpnet model offer superior ceiling performance. Second, we validate the effectiveness of MRL through extensive saturation and retention analysis. Third, we perform a qualitative analysis of downstream RAG performance, correlating retrieval metrics with generation faithfulness.

2 Related Work

2.1 Retrieval-Augmented Generation (RAG)

The core concept of augmenting generation with external knowledge, pioneered by [1], has evolved into a complex and rapidly advancing subfield of NLP. The academic discourse has broadly categorized its evolution into distinct phases. Early implementations, often referred to as “Naive RAG,” employed a straightforward retrieve-then-read pipeline where a user query directly drives a semantic search, and the top-k retrieved documents are concatenated into the LLM’s context window. While effective, this approach is susceptible to issues like misaligned retrieval and context overload [3].

Recent literature details a shift towards “Advanced RAG,” which incorporates sophisticated pre- and post-retrieval strategies. These include query transformations (e.g., query rewriting, expansion), advanced indexing techniques (e.g., multi-vector retrieval), and re-ranking mechanisms to refine retrieved results before generation [2]. The current state-of-the-art is moving towards “Modular RAG,” where the pipeline is deconstructed into interconnected modules that can be dynamically orchestrated based on query complexity, showcasing the field’s progression towards more robust and adaptive systems. Our work focuses on the foundational retrieval module, as its optimization is a prerequisite for the success of any advanced RAG strategy, particularly in non-English contexts.

2.2 Semantic Search in Low-Resource Languages

While multilingual models offer a zero-shot baseline for many languages, their performance in specialized retrieval tasks for low-resource languages is often suboptimal. This performance gap stems from several factors, including

inefficient tokenization of morphologically rich languages, semantic drift where word meanings are biased towards high-resource training data, and a lack of cultural and domain-specific context [4]. Recent work has begun to tackle these challenges directly within the Bengali language ecosystem. For example, [5] explored fine-tuning a generative model, Llama, for Bengali question-answering, establishing new benchmarks. However, such generative approaches are fundamentally dependent on the quality of retrieved context. An inaccurate or irrelevant context will invariably lead to a poor or hallucinated response, regardless of the generator’s quality.

This dependency underscores the criticality of a robust retrieval component. Addressing this challenge has become a critical area of research. For instance, [4] demonstrates that fine-tuning multilingual models on even small, curated datasets in the target language can yield substantial performance gains. Other innovative approaches aim to overcome data scarcity through novel training paradigms. A notable example is Retrieval-Augmented Retrieval (RAR) [6], which uses a generate-then-retrieve cycle to synthesize training data, a technique that holds promise for languages where labeled pairs are scarce. Our study contributes to this area by providing a rigorous empirical benchmark that quantifies the impact of task-specific fine-tuning on diverse model architectures for Bengali, offering a clear pathway for practitioners to move beyond unreliable zero-shot baselines.

2.3 Advancements in Sentence Embeddings for Bengali

The Bengali NLP landscape has matured significantly with the development of foundational models. Early efforts produced models like BanglaBERT [11], which was pre-trained on a large Bengali corpus using a Masked Language Modeling (MLM) objective. While powerful for natural language understanding tasks, the pooling of raw token embeddings from such models is not inherently optimized for semantic similarity tasks. This limitation spurred the development of dedicated sentence-transformer models.

The advent of bi-encoder architectures, popularized by Sentence-BERT (SBERT) [10], marked a paradigm shift. This methodology has been actively adopted for Bengali, with a common strategy being cross-lingual knowledge distillation from powerful English teacher models. For instance, the shihab17/bangla-sentence-transformer used in our study was developed by fine-tuning an XLM-RoBERTa student model on machine-translated datasets, using a multilingual SBERT model as the teacher [12]. Similarly, the BanglaEmbed models were created using a comparable distillation technique to produce lightweight and efficient sentence encoders [14]. Our work builds directly upon this foundation. By applying a unified, task-specific fine-tuning regimen to these and other architectures, we aim to move beyond the initial pre-training and isolate the impact of the base model’s architectural priors on final retrieval performance for a demanding question-answering task.

2.4 Efficient Representation Learning

As vector databases grow to billions of entries, the dimensionality of embeddings becomes a critical factor for storage costs, memory usage, and query latency. Matryoshka Representation Learning (MRL) [9] offers a principled solution to this challenge.

Formally, MRL reframes representation learning as a multi-scale optimization problem. The central idea is to learn a high-dimensional embedding vector, $z \in \mathbb{R}^d$, such that its prefixes are also effective, lower-dimensional representations. To achieve this, MRL defines a set of nested “Matryoshka” dimensions, $M \subset \{1, \dots, d\}$, typically chosen by halving the dimensionality (e.g., $M = \{d, d/2, d/4, \dots, 64\}$). The model, parameterized by θ_F , learns a function $F(\cdot; \theta_F)$ that maps an input x to the embedding z .

The key innovation lies in modifying the training objective. Instead of calculating the loss only on the final d -dimensional embedding, MRL computes and aggregates the loss at each of the chosen granularities $m \in M$. For a given training objective $\mathcal{L}_{\text{task}}$, such as the Multiple Negatives Ranking Loss used in our work, the MRL objective for a single data point (x_i, y_i) is formulated as a weighted sum over the nested dimensions:

$$\min_{\theta_F, \{W^{(m)}\}_{m \in M}} \frac{1}{N} \sum_{i=1}^N \sum_{m \in M} c_m \cdot \mathcal{L}_{\text{task}} \left(W^{(m)} \cdot F(x_i; \theta_F)_{1:m}, y_i \right) \quad (1)$$

Here, $F(x_i; \theta_F)_{1:m}$ denotes the output embedding truncated to its first m dimensions, $W^{(m)}$ is a task-specific classification head for that dimension, and c_m are scaling coefficients (typically set to 1). By optimizing this composite loss, the model is explicitly forced to encode the most critical semantic information into the initial dimensions of the embedding vector, with finer details layered into subsequent dimensions.

This approach stands in contrast to traditional dimensionality reduction techniques such as Principal Component Analysis (PCA) or autoencoders, which require a separate, post-hoc training step and can result in significant information loss. MRL’s primary advantage is that it integrates efficiency directly into the training objective, enabling flexible, on-the-fly truncation at inference time. An application can use the full 768-dimensional vector for high-precision tasks and a truncated 128-dimensional prefix for low-latency lookups, all from a single learned representation. Our study is the first to systematically apply and evaluate MRL for the Bengali language, demonstrating its viability for creating highly efficient yet powerful retrieval systems.

3 Methodology

Our experimental framework involves fine-tuning three distinct base models using a unified training strategy to ensure a fair comparison. See the figure 1.

3.1 Base Models

We selected three models representing different points on the efficiency-performance spectrum.

shihab17/bangla-sentence-transformer: A monolingual model based on the XLM-R architecture, created via multilingual knowledge distillation [12]. It represents the specialized Bengali approach, producing 768-dimensional vectors.

distiluse-base-multilingual-cased-v1: A knowledge-distilled version of the Universal Sentence Encoder [13]. It produces 512-dimensional vectors and is designed for high-throughput, low-latency applications.

paraphrase-multilingual-mpnet-base-v2: A model based on the MPNet architecture, trained on a massive multilingual corpus of paraphrase pairs [13]. It is generally regarded as having stronger semantic modeling capabilities than standard BERT or RoBERTa models.

3.2 Dataset Preparation

We utilized the BanglaRQA dataset [8], comprising human-annotated questions and answers from Bengali Wikipedia. We combined the training and validation splits to create a training corpus of 8,573 positive pairs. Crucially, we filtered for answerable questions to ensure high-quality positive signals. The test set consisted of 1,405 held-out pairs.

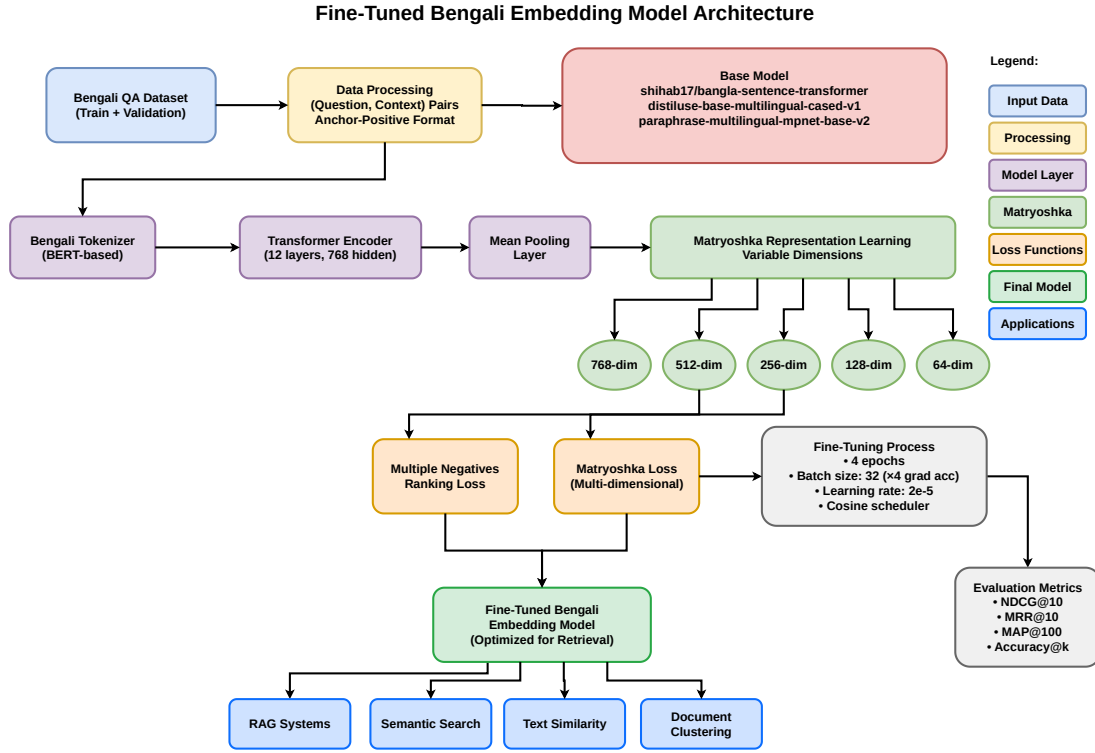


Fig. 1. The comparative fine-tuning architecture. Three distinct base models are trained using a shared pipeline involving Siamese networks, Mean Pooling, and a Matryoshka-wrapped Multiple Negatives Ranking Loss function.

Table 1. BanglaRQA Dataset Samples

Pass ID	Title	Context (Truncated)	Q. ID	Question	Ans?	Type	Answers
bn_wiki_0981	ইন্টেল কর্পোরেশন	২০০০ সালের পর, প্রসেসরের চাহিদা হ্রাস পায়। প্রতিযোগিতা, বিশেষ করে এএমডি শেয়ার বাজারের বড় অংশ অধিগ্রহণ করে। শুরুতে কম ক্ষমতার এবং মধ্যম ক্ষমতার প্রসেসর এবং ধীরে ধীরে পুরো পণ্যের বাজার এবং মূল বাজারে ইন্টেলের অবস্থান আশঙ্কাজনকহারে কমে যায়। [...]	bn_wiki_0981_01	কত সালের পর, প্রসেসরের চাহিদা হ্রাস পায়?	1	factoid	{ "text": ["২০০০", "২০০০"], "type": ["single span", "single span"] }
bn_wiki_0656	উবুন্টু (লিনাক্স)	উবুন্টু ইনস্টল করার জন্য সাধারণভাবে লাইভ সিডি ব্যবহার করা হয়। উবুন্টু অপারেটিং সিস্টেমটি ইনস্টল করা ছাড়াই সরাসরি সিডি থেকে ব্যবহার করা যায়। তবে লাইভ সিডি ব্যবহার করা হলে পূর্ণাঙ্গ অপারেটিং সিস্টেমের অনেক সুবিধাই এখানে পাওয়া যাবে না। [...]	bn_wiki_0656_02	কি নামে উবুন্টু ওএস ইনস্টল করতে সফটওয়্যার ব্যবহার করা হয়?	0	factoid	{ "text": ["", ""], "type": ["", ""] }

3.3 Fine-tuning Strategy

We employed a Siamese network structure with shared weights. The training objective was a summation of Multiple Negatives Ranking Loss (MNRL) calculated across varying dimensions, facilitated by Matryoshka Representation Learning.

MNRL treats the corresponding context for a question as the positive sample and all other contexts in the batch as negatives. The Matryoshka Loss wraps this objective, calculating the loss at dimensions $\{d_{max}, \dots, 64\}$. This forces the model to encode the most salient semantic information into the first 64 dimensions.

4 Quantitative Evaluation

We evaluated the models on the BanglaRQA test set using standard Information Retrieval metrics. Below we define the key metrics used in our analysis: Accuracy@k, Mean Reciprocal Rank (MRR), Mean Average Precision (MAP), and Normalized Discounted Cumulative Gain (NDCG). Let Q be the set of all queries in the test set.

- **Accuracy@k:** This metric measures the proportion of queries for which at least one correct document is found within the top k retrieved results. It is a simple measure of success within a fixed retrieval window.
- **Mean Reciprocal Rank (MRR):** MRR evaluates how quickly the system finds the first relevant document. For each query, the reciprocal rank is the inverse of the rank of the first correct answer ($1/\text{rank}_i$). If no correct document is retrieved, the reciprocal rank is 0. The MRR is the average of these values across all queries.

$$\text{MRR} = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \frac{1}{\text{rank}_i} \quad (2)$$

- **Mean Average Precision (MAP):** MAP provides a single-figure measure of quality across recall levels. For a single query q , Average Precision (AP) is the average of precision values calculated at each point a relevant document is found in the ranked list. MAP is the mean of these AP scores over all queries.

$$\text{AP}(q) = \frac{\sum_{k=1}^n (P(k) \times \text{rel}(k))}{\text{number of relevant documents for } q} \quad (3)$$

$$\text{MAP} = \frac{1}{|Q|} \sum_{q \in Q} \text{AP}(q) \quad (4)$$

where n is the number of retrieved documents, $P(k)$ is the precision at cutoff k , and $\text{rel}(k)$ is an indicator function that is 1 if the item at rank k is relevant, and 0 otherwise.

- **Normalized Discounted Cumulative Gain (NDCG):** NDCG evaluates the quality of a ranking by considering the graded relevance of documents and penalizing relevant documents that appear lower in the list. It is calculated by dividing the Discounted Cumulative Gain (DCG) by the Ideal DCG (IDCG), which represents the gain of a perfect ranking. This normalization ensures the score is between 0 and 1.

$$\text{DCG}_k = \sum_{i=1}^k \frac{\text{rel}_i}{\log_2(i+1)} \quad (5)$$

$$\text{NDCG}_k = \frac{\text{DCG}_k}{\text{IDCG}_k} \quad (6)$$

where rel_i is the relevance score of the document at position i (in our case, 1 for relevant, 0 for not).

4.1 Performance Comparison

The comprehensive results of our experiments are visualized in Figure 2. The mpnet-base model (green bars) emerged as the clear leader across all four metrics. Even at lower dimensions, it consistently outperformed the monolingual shihab17 (blue) and distilled distiluse (red) models.

Table 2 details the specific improvements. The distiluse model, while efficient, started with a negligible baseline performance (NDCG@10 of 0.0037). Fine-tuning resurrected this model, bringing it to a respectable NDCG@10 of 0.4213. The monolingual shihab17 model showed strong gains, improving to 0.5638. However, mpnet achieved the highest scores, with an NDCG@10 of 0.6139 and an Accuracy@1 of 54.36%.

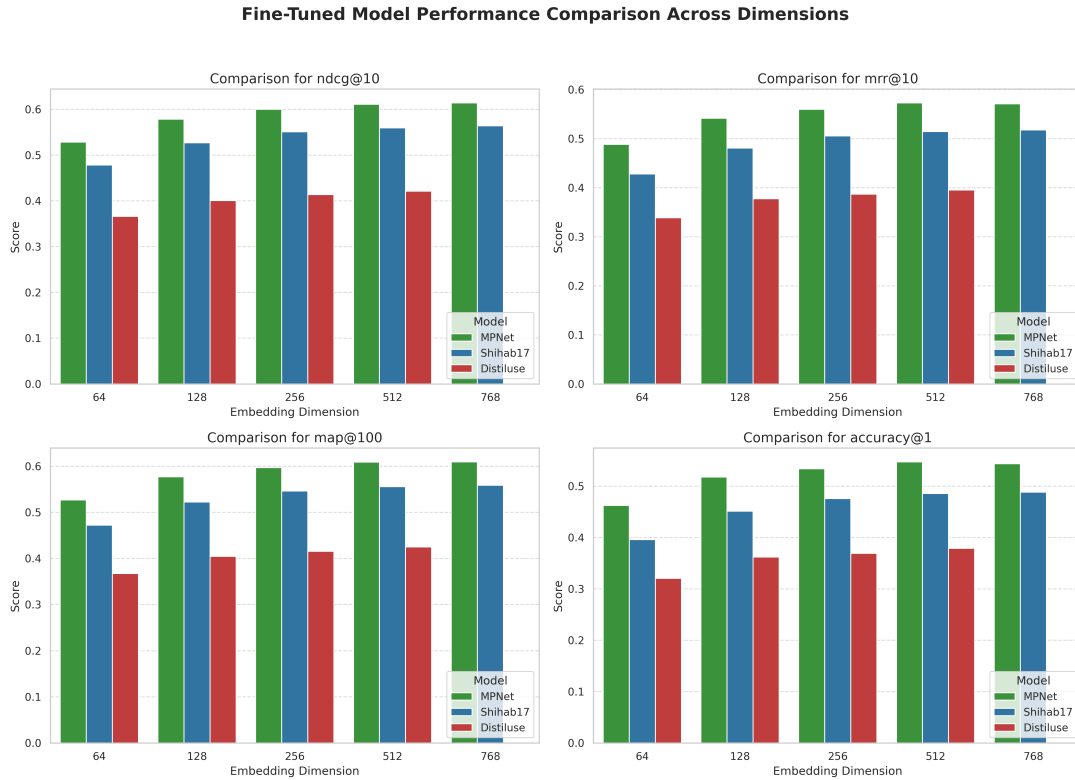


Fig. 2. Fine-Tuned Model Performance Comparison Across Dimensions. The MPNet model (green) demonstrates superior performance across NDCG@10, MRR@10, MAP@100, and Accuracy@1, maintaining its lead even as embedding dimensions are reduced.

To rigorously evaluate the efficacy of our fine-tuning approach, we conducted a comparative analysis against the pre-trained base models across varying vector dimensions ranging from 64d to 768d. As illustrated in Fig. 3, the fine-tuned models (indicated by the red squares) consistently outperform their base counterparts (blue circles) by a significant margin across all three architectures.

In terms of ranking quality (NDCG@10), the MPNet and Shihab17 (BanglaBERT) models show substantial gains. For instance, the fine-tuned MPNet achieves an NDCG@10 score of 0.6139 at 768 dimensions, compared to 0.2695 for the base model—an improvement of approximately 127%.

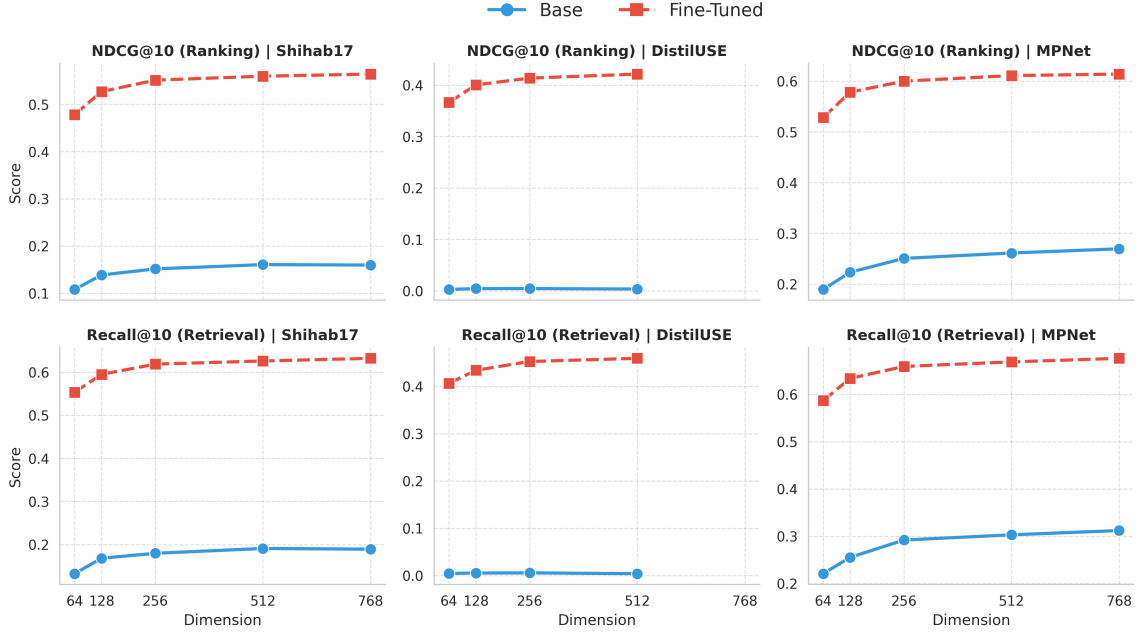


Fig. 3. **Impact of fine-tuning on semantic search performance across varying dimensions (64d–768d).** The comparison highlights the efficacy gap between Base (Blue/Circle) and Fine-Tuned (Red/Square) models. The top row displays Ranking Quality (NDCG@10), while the bottom row displays Retrieval Coverage (Recall@10). Fine-tuned models demonstrate superior performance and robustness to dimensionality reduction compared to pre-trained base models.

Most notably, the DistilUSE base model exhibited near-zero performance (< 0.01 across all metrics), indicating a lack of semantic alignment for the target language in its pre-trained state. However, post-fine-tuning, the model recovered to a competitive NDCG@10 of 0.4213, demonstrating that our training methodology successfully aligned the multilingual vector space.

Furthermore, Fig. 3 demonstrates the robustness of the fine-tuned models to dimensionality reduction. While performance naturally degrades as dimensions are reduced towards 64d, the fine-tuned models maintain high utility. Specifically, the fine-tuned Shihab17 model at 64d retains an NDCG score of 0.4783, which surpasses the performance of the full 768d base MPNet model. This suggests that our fine-tuned models are highly effective for resource-constrained environments where low-dimensional embeddings are required for storage efficiency.

Table 2. Comparative Performance of Fine-Tuned (FT) Models vs. Baselines at Maximum Dimensions. The best scores are highlighted in **bold**. The $\dagger$ symbol indicates that the best model (MPNet) is statistically significantly better than the second-best model (Shihab17) with $p < 0.001$.

Model	Dim	NDCG@10		MRR@10		MAP@100		Accuracy@10		Precision@10		Recall@10	
		Base	FT	Base	FT	Base	FT	Base	FT	Base	FT	Base	FT
DistilUSE	512d	0.0037	0.4213	0.0036	0.3955	0.0056	0.4248	0.0044	0.4698	0.0014	0.1770	0.0040	0.4592
Shihab17	768d	0.1599	0.5638	0.1411	0.5179	0.1651	0.5586	0.1993	0.6557	0.0715	0.2424	0.1889	0.6323
MPNet	768d	0.2695	0.6139$\dagger$	0.2413	0.5712$\dagger$	0.2761	0.6094$\dagger$	0.3283	0.6913$\dagger$	0.1201	0.2593$\dagger$	0.3120	0.6762$\dagger$

Table 2 presents a detailed breakdown of the results at the maximum embedding dimensions available for each architecture. Across all metrics, fine-tuning delivered clear performance gains.

Most strikingly, the DistilUSE model moved from an almost unusable baseline (NDCG of 0.0037) to a strong post-fine-tuning score of 0.4213. MPNet reached the best overall performance, with an Accuracy@10 of 0.6913—meaning the correct item appeared in the top ten nearly 70% of the time, up from 32% in the base version. The Shihab17 model also saw a three-fold jump in Recall@10, rising from 0.1889 to 0.6323.

4.2 Statistical Significance Analysis

To ascertain that the performance superiority of the mpnet architecture is not merely an artifact of specific query variations or random initialization, we conducted a statistical significance test on the test set predictions. Given that information retrieval metrics such as NDCG are typically non-normally distributed, we employed the **Wilcoxon Signed-Rank Test**, a non-parametric paired difference test.

We formulated the null hypothesis (H_0) that there is no significant difference in the median retrieval performance between the best-performing model (mpnet-ft) and the runner-up models. We compared the per-query NDCG@10 scores across the 1,405 test queries.

The results, summarized in Table 3, strongly reject the null hypothesis. The fine-tuned mpnet model demonstrates a statistically significant improvement over the monolingual shihab17 model ($p < 0.001$) and the distilled distiluse model ($p < 0.001$). The extremely low p-values (e.g., 1.0×10^{-47} against DistilUSE) confirm that mpnet’s architectural advantages translate into robust, consistent gains across the vast majority of the test distribution.

Table 3. Wilcoxon Signed-Rank Test Results (NDCG@10)

Model A	Model B	Statistic	p-value
MPNet-FT	Shihab17-FT	8.02×10^3	7.81×10^{-6}
MPNet-FT	DistilUSE-FT	8.30×10^3	1.00×10^{-47}
Shihab17-FT	DistilUSE-FT	1.12×10^4	5.35×10^{-38}

4.3 Matryoshka Efficiency and Saturation

A critical analysis of the results at lower dimensions highlights the success of the Matryoshka training strategy. As illustrated in the saturation curves in Figure 4, all models exhibit a logarithmic growth in performance as dimensions increase, eventually reaching a plateau. The mpnet model (green line) consistently dominates across all dimensions. Notably, the “elbow” of the curve appears early; moving from 256 to 768 dimensions yields diminishing returns. Figure 5 further illustrates this saturation effect from a classification metric perspective, showing that individual scores for accuracy, f1, precision, and recall follow a similar logarithmic growth pattern, with most gains realized by the 512-dimension mark.

To quantify this efficiency, we analyzed the performance retention relative to the best-performing MPNet configuration (Figure 6). The results are striking: MPNet achieves over 95% of its maximum capacity at just 128 dimensions. This implies that for large-scale Bengali retrieval systems, one can achieve state-of-the-art accuracy with a fraction of the storage usually necessary.

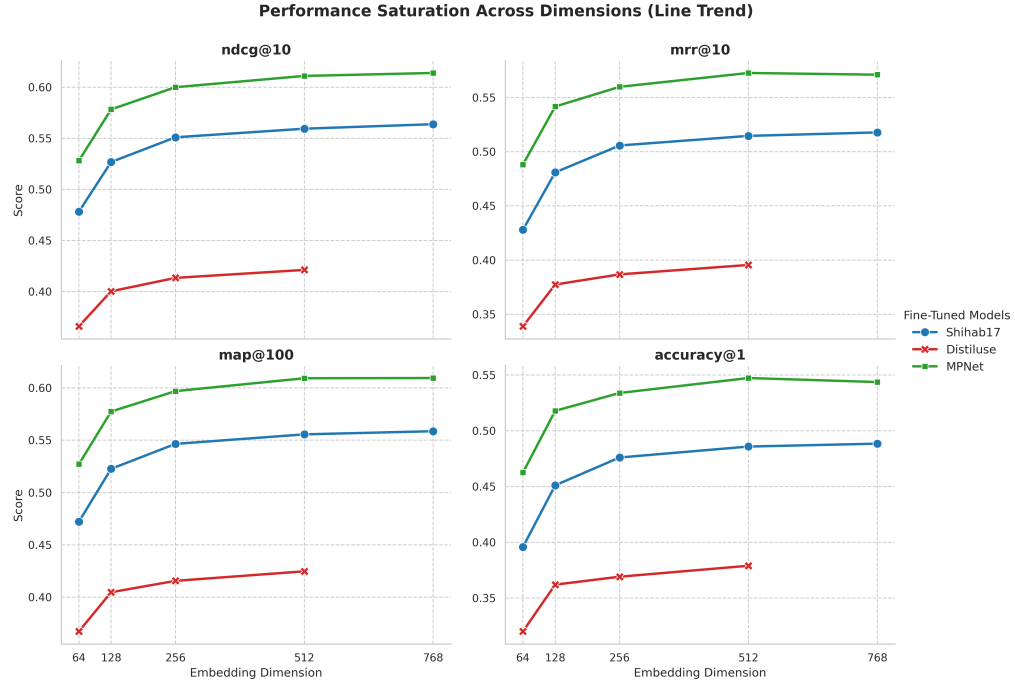


Fig. 4. Performance saturation across dimensions for different metrics. The MPNet model demonstrates superior scaling, reaching near-peak performance earlier than the monolingual or distilled baselines.

4.4 Generalization on MTEB Benchmarks

To assess the generalization capabilities of our fine-tuned models beyond the primary question-answering task, we evaluated them on a selection of Bengali tasks from the Massive Text Embedding Benchmark (MTEB). The selected tasks included ‘BengaliDocumentClassification.v2’ (BDC), ‘BengaliHateSpeechClassification.v2’ (BHSC), and the ‘Tatoeba’ bitext retrieval task. This evaluation measures how well the semantic space learned on BanglaRQA transfers to broader classification and retrieval contexts.

The results, depicted in Figure 7, reinforce the superiority of the fine-tuned MPNet architecture. The model-wise breakdown clearly shows that MPNet consistently achieves the highest scores across all three datasets and nearly all evaluation metrics. While the monolingual ‘shihab17’ model shows competitive performance, particularly on the classification tasks (BDC, BHSC), it does not reach the overall proficiency of MPNet. The ‘distiluse’ model lags significantly behind both. This broader evaluation confirms that the architectural advantages of MPNet, enhanced by our fine-tuning strategy, translate into robust, general-purpose semantic representations for the Bengali language.

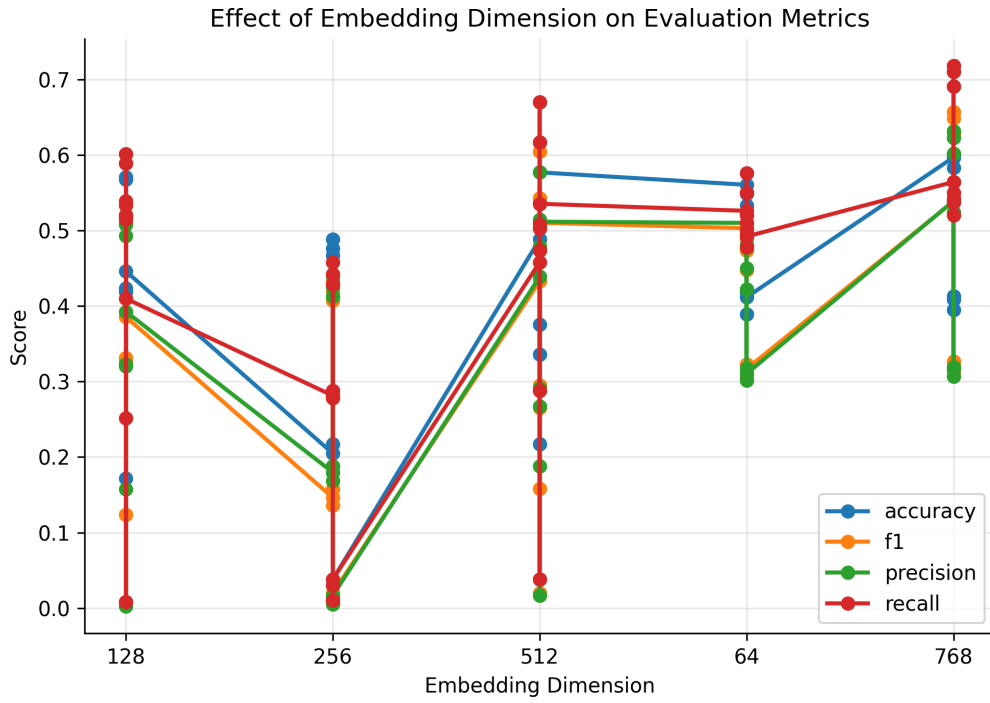


Fig. 5. Effect of Embedding Dimension on Evaluation Metrics for the fine-tuned MPNet model. Performance across key metrics begins to plateau after 256-512 dimensions, reinforcing the effectiveness of Matryoshka Representation Learning.

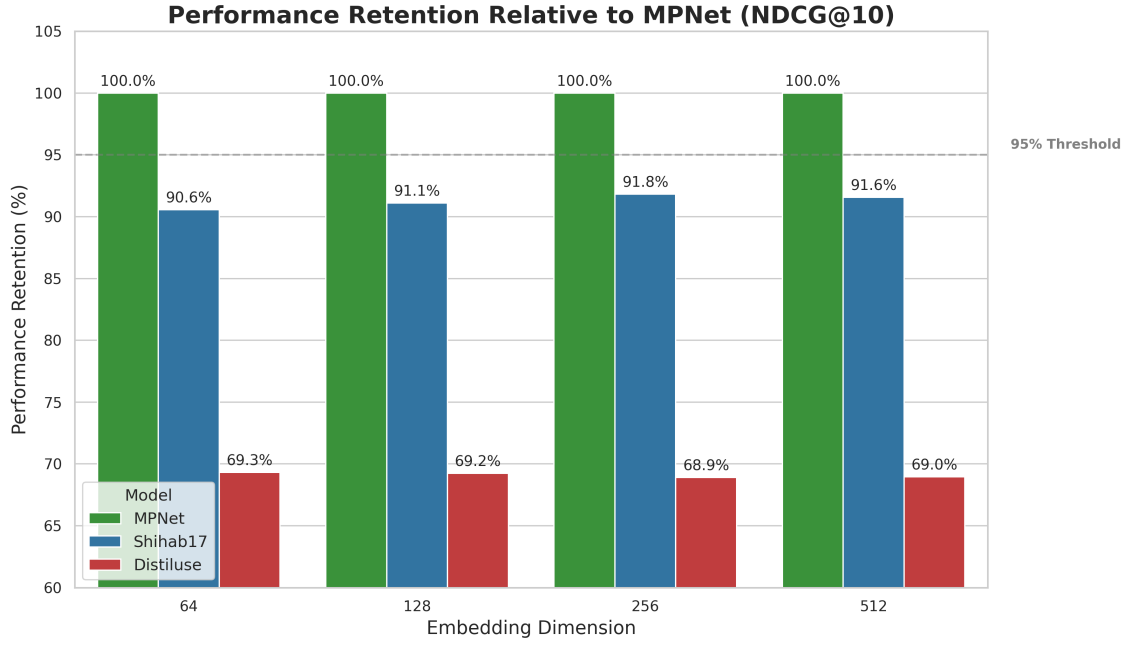


Fig. 6. Performance Retention Analysis (NDCG@10). Bars indicate the percentage of performance retained relative to the full-precision MPNet model. The dashed line indicates the 95% efficiency threshold.

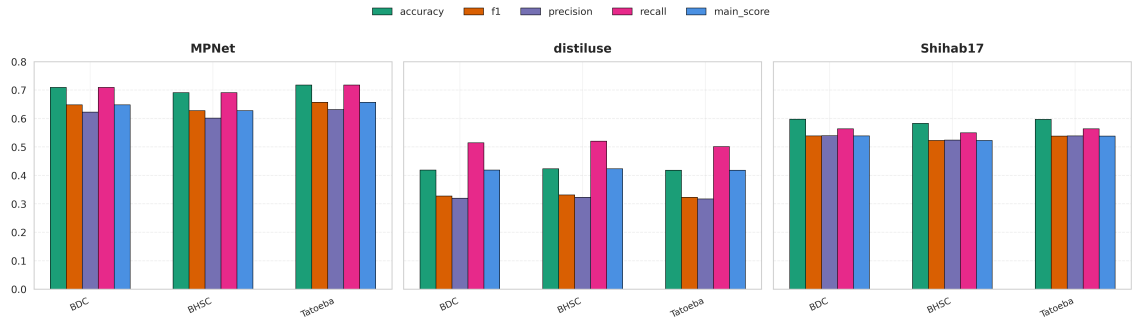


Fig. 7. Comparative performance on MTEB benchmark tasks, broken down by model. Each subplot shows a model's performance on Bengali Document Classification (BDC), Bengali Hate Speech Classification (BHSC), and Tatoeba retrieval. The MPNet model demonstrates consistently superior scores across all tasks and metrics.

5 Qualitative RAG Analysis

To understand the downstream impact of these retrieval metrics, we integrated the fine-tuned models into a RAG pipeline. We utilized Radar Charts (Figure 8) to visualize the RAG metrics (Faithfulness, Answer Relevancy, Context Precision) comparing the low-resource (64d) and high-resource (512d) configurations.

Table 4. Comparative RAG Responses: Success vs. Failure Cases

Query	Model	Retrieved-Context	Generated Response	Result
বিশ্বের প্রথম কম্পিউটার কে তৈরি করেন ? (Who created the first computer?)	mpnet-ft	['১৮৩৭ খ্রীষ্টাব্দে চার্লস ব্যাবেজ একটি বিশ্লেষণধর্মী বা অ্যানালিটিক্যাল ইঞ্জিন-এর খসড়া তৈরি করেন যার মূল সুবিধাগুলি ছিল - এক্সপ্যান্ডেবল মেমরি (বিস্তারযোগ্য যান্ত্রিক স্মৃতিশক্তি), গাণিতিক হিসাব করার ক্ষমতা, যুক্তিনির্ভর সিদ্ধান্ত নেওয়ার ক্ষমতা, লুপ ও কন্ডিশনাল ব্রাঞ্চিং(যা আধুনিক প্রোগ্রামিং ভাষাগুলিতে পাওয়া যায়) ইত্যাদি। এই যন্ত্রটিকে বিজ্ঞানীরা বিশ্বের প্রথম কম্পিউটার বলে স্বীকার করে থাকেন এবং টুরিংসমতুল্য বলেও মনে করা হয়।', 'কম্পিউটারের গঠন ও প্রচলন নীতির ভিত্তিতে একে তিন ভাগে ভাগ করা যায়।', 'উইন্ডোজে ইউনিকোড ভিত্তিক বাংলা লেখার জন্য ২০০০ সালের ২৬শে মার্চ অত্র কীবোর্ড সফটওয়্যারটি আবির্ভূত হয়[1][2]। এটার সাহায্যে বাংলা লিপি ব্যবহার করে এমন সব ভাষাতেই টাইপ করা যায়। এ ধরনের ভাষার মধ্যে অসমীয়া ভাষা অন্যতম। মেহদী হাসান খান নামে ময়মনসিংহ মেডিকেল কলেজের একজন ছাত্র ২০০৩ সালে অত্র কীবোর্ড তৈরির কাজ শুরু করেন। তিনি এটি সর্বপ্রথম তৈরি করেছিলেন ডিজিটাল বেসিক প্রোগ্রামিং ভাষা দিয়ে, পরবর্তীতে তিনি তা ডেলফিতে ভাষান্তর করেন। এই সফটওয়্যারটির লিনাক্স সংস্করণ লেখা হয়েছে সি++ প্রোগ্রামিং']	বিশ্বের প্রথম কম্পিউটার চার্লস ব্যাবেজ তৈরি করেন।	Success
বিশ্বের প্রথম কম্পিউটার কে তৈরি করেন ? (Who created the first computer?)	distiluse-ft	['কম্পিউটারের গঠন ও প্রচলন নীতির ভিত্তিতে একে তিন ভাগে ভাগ করা যায়।', 'লুমিয়ের সিনেমাটোগ্রাফ'-এর মত বিশ্বব্যাপী প্রভাব বিস্তার করতে পারেনি। এই ঘটনার পরে পরেই সারা বিশ্ব জুড়ে প্রচুর ফিল্ম প্রোডাকশন কোম্পানি তৈরী হয়ে যায় আর মাত্র এক দশকের মধ্যেই চলচ্চিত্র তার অভিনবত্ব কাটিয়ে উঠে এক বিশাল সর্বজনীন বিনোদনশিল্পে পরিণত হয়।', 'লিনাক্স সংস্করণ লেখা হয়েছে সি++ প্রোগ্রামিং ভাষায়। পরবর্তীতে রিফাত-উন-নবী, তানবিন ইসলাম সিয়াম, রাইয়ান কামাল, শাবাব মুস্তফা, ওমর ওসমান, সারিম খান এবং নিপুন হক এই সফটওয়্যারের উন্নয়নের সাথে যুক্ত হন।[1][3]']	আমি জানি না।	Failure
ঔপন্যাসিক ও গল্পকার জহির রায়হান পরিচালিত প্রথম চলচ্চিত্রের নাম কী ? (Name of first film by Jahir Raihan?)	mpnet-ft	['জহির রায়হান বাংলা সাহিত্যে স্নাতকোত্তর ডিগ্রি লাভ করেন। তার সাহিত্যিক ও সাংবাদিক জীবন শুরু হয়। ১৯৫০ সালে তিনি যুগের আলো পত্রিকায় সাংবাদিক হিসেবে কাজ করা শুরু করেন। পরবর্তীতে তিনি খাপছাড়া, যান্ত্রিক, সিনেমা ইত্যাদি পত্রিকাতেও কাজ করেন। ১৯৫৬ সালে তিনি সম্পাদক হিসেবে প্রবাহ পত্রিকায় যোগ দেন। ১৯৫৫ সালে তার প্রথম গল্পগ্রন্থ সূর্যগ্রহণ প্রকাশিত হয়। চলচ্চিত্র জগতে তার পদার্পণ ঘটে ১৯৫৭ সালে, জাগো হুয়া সাভেরা ছবিতে সহকারী হিসেবে কাজ করার মাধ্যমে। তিনি সালাউদ্দীনের ছবি যে নদী মরুপথে</i>তেও সহকারী', 'তিনি সালাউদ্দীনের ছবি যে নদী মরুপথে</i>তেও সহকারী হিসেবে কাজ করেন। প্রখ্যাত চলচ্চিত্র পরিচালক এহতেশাম তাকে এ দেশ তোমার আমার এ কাজ করার আমন্ত্রণ জানান; জহির এ ছবির নামসঙ্গীত রচনা করেছিলেন। ১৯৬১ সালে তিনি রূপালী জগতে পরিচালক হিসেবে আত্মপ্রকাশ করেন কখনো আসেনি চলচ্চিত্রের মাধ্যমে। ১৯৬৪ সালে তিনি পাকিস্তানের প্রথম রঙিন চলচ্চিত্র সঙ্গম নির্মাণ করেন (উর্দু ভাষার ছবি) এবং পরের বছর তার প্রথম সিনেমাস্কোপ চলচ্চিত্র বাহানা মুক্তি দেন। জহির রায়হান ভাষা আন্দোলনে সক্রিয়ভাবে যুক্ত ছিলেন এবং ২১শে ফেব্রুয়ারির', 'বেশ কয়েকটি প্রদর্শনী হয় এবং চলচ্চিত্রটি দেখে সত্যজিত রায়, মৃণাল সেন, তপন সিনহা এবং ঋদ্ধিক ঘটক প্রমুখ ভূয়সী প্রশংসা করেন। সে সময়ে তিনি চরম অর্থনৈতিক দৈন্যের মধ্যে থাকা সত্ত্বেও তার চলচ্চিত্র প্রদর্শনী হতে প্রাপ্ত সমুদয় অর্থ তিনি মুক্তিযোদ্ধা তহবিলে দান করে দেন।[2]']	জহির রায়হান পরিচালিত প্রথম চলচ্চিত্রের নাম "কখনো আসেনি"	Success

Continued on next page...

Table 4. Comparative RAG Responses (Continued)

Query	Model	Retrieved-Context	Generated Response	Result
উপন্যাসিক ও গল্পকার জহির রায়হান পরিচালিত প্রথম চলচ্চিত্রের নাম কী ? (Name of first film by Jahir Raihan?)	shihab17-ft	[জহির রায়হান বাংলা সাহিত্যে স্নাতকোত্তর ডিগ্রি লাভ করেন। তার সাহিত্যিক ও সাংবাদিক জীবন শুরু হয়। ১৯৫০ সালে তিনি যুগের আলো পত্রিকায় সাংবাদিক হিসেবে কাজ করা শুরু করেন। পরবর্তীতে তিনি খাপছাড়া, যান্ত্রিক, সিনেমা ইত্যাদি পত্রিকাতেও কাজ করেন। ১৯৫৬ সালে তিনি সম্পাদক হিসেবে প্রবাহ পত্রিকায় যোগ দেন। ১৯৫৫ সালে তার প্রথম গল্পগ্রন্থ সূর্যগ্রহণ প্রকাশিত হয়। চলচ্চিত্র জগতে তার পদার্পণ ঘটে ১৯৫৭ সালে, জাগো হুয়া সাভেরা ছবিতে সহকারী হিসেবে কাজ করার মাধ্যমে। তিনি সালাউদ্দীনের ছবি যে নদী মরুপথে</i>তেও সহকারী, দৈবশ্রী রায় এর জন্ম, বেড়ে ওঠা কলকাতায়। তার পিতার নাম বীরেন্দ্র কিশোর রায় এবং মায়ের নাম আরতি রায়। ১৯৬৯ সনে তিনি প্রথম "বালক গদাধর" নামে একটি চলচ্চিত্রে শিশুশিল্পী হিসেবে অভিনয় করেন, যার পরিচালক ছিলেন হরিময় সেন। তিনি কলকাতার এ পি জে ,পার্ক স্ট্রীট শাখার স্কুলের ছাত্রী ছিলেন, যদিও নাচের প্রতি অত্যধিক ঝোঁক থাকার কারণে বেশিদূর এগোতে পারেন নি। তার ডাক নাম চুমকি, বৈশিষ্ট্য কয়েকটি প্রদর্শনী হয় এবং চলচ্চিত্রটি দেখে সত্যজিত রায়, মুগাল সেন, তপন সিনহা এবং ঋত্বিক ঘটক প্রমুখ ভূয়সী প্রশংসা করেন। সে সময়ে তিনি চরম অর্থনৈতিক দৈন্যের মধ্যে থাকা সত্ত্বেও তার চলচ্চিত্র প্রদর্শনী হতে প্রাপ্ত সমুদয় অর্থ তিনি মুক্তিযোদ্ধা তহবিলে দান করে দেন।[2]]	আমি জানি না।	Failure
ভারতের ছত্তীসগড় রাজ্যের রাজধানী কোথায় ? (Where is the capital of Chhattisgarh?)	mpnet-ft	[ছত্তীসগড় (ছত্তীসগড়ি/হিন্দি: মধ্য ভারতের একটি রাজ্য। ২০০১ সালের ১ নভেম্বর মধ্যপ্রদেশ রাজ্যের দক্ষিণ-পশ্চিমাঞ্চলে স্থিত ষোলোটি ছত্তীসগড়ি-ভাষী জেলা নিয়ে এই রাজ্য স্থাপন করা হয়। রায়পুর এই রাজ্যের রাজধানী। ৫২,১৯৯ বর্গমাইল (১৩৫,১৯৪ বর্গকিলোমিটার) আয়তন বিশিষ্ট এই রাজ্য ভারতের দশম বৃহত্তম রাজ্য।, দৈবশ্রী রায় এর জন্ম, বেড়ে ওঠা কলকাতায়। তার পিতার নাম বীরেন্দ্র কিশোর রায় এবং মায়ের নাম আরতি রায়। ১৯৬৯ সনে তিনি প্রথম "বালক গদাধর" নামে একটি চলচ্চিত্রে শিশুশিল্পী হিসেবে অভিনয় করেন, যার পরিচালক ছিলেন হরিময় সেন। তিনি কলকাতার এ পি জে ,পার্ক স্ট্রীট শাখার স্কুলের ছাত্রী ছিলেন, যদিও নাচের প্রতি অত্যধিক ঝোঁক থাকার কারণে বেশিদূর এগোতে পারেন নি। তার ডাক নাম চুমকি, জমিদারি নিলামে ক্রয় করেন। তাঁর জমিদারিতে বেশ কিছু অংশীদার ও সহ-অংশীদার ছিলেন। কিন্তু তিনি বহরমপুর, পাণ্ডুয়া, কালীগ্রাম ও শাহজাদপুরে চারটি বড় জমিদারির মালিক ছিলেন এবং এগুলিতে তাঁর কোন অংশীদার ছিল না। ১৮৪০ সালে সেগুলি তিনি তাঁর সন্তান ও তাঁদের উত্তরাধিকারীদের ট্রাস্ট করে দেন। দ্বারকানাথ ঠাকুরের জমিদারি পরিচালনার একটি বিশেষ বৈশিষ্ট্য ছিল। তিনি এটিকে সামন্ততন্ত্রের দৃষ্টিকোণ থেকে না দেখে মূলধনের সৃষ্টিশীল প্রসার হিসেবে বিবেচনা করেন। এক্ষেত্রে তিনি সমকালীন জমিদারদের চেয়ে ব্যতিক্রম ছিলেন। জমিদারি]	রায়পুর।	Success
ভারতের ছত্তীসগড় রাজ্যের রাজধানী কোথায় ? (Where is the capital of Chhattisgarh?)	distiluse-ft	[ছত্তীসগড় (ছত্তীসগড়ি/হিন্দি: মধ্য ভারতের একটি রাজ্য। ২০০১ সালের ১ নভেম্বর মধ্যপ্রদেশ রাজ্যের দক্ষিণ-পশ্চিমাঞ্চলে স্থিত ষোলোটি ছত্তীসগড়ি-ভাষী জেলা নিয়ে এই রাজ্য স্থাপন করা হয়। রায়পুর এই রাজ্যের রাজধানী। ৫২,১৯৯ বর্গমাইল (১৩৫,১৯৪ বর্গকিলোমিটার) আয়তন বিশিষ্ট এই রাজ্য ভারতের দশম বৃহত্তম রাজ্য।, দৈবশ্রী রায় এর জন্ম, বেড়ে ওঠা কলকাতায়। তার পিতার নাম বীরেন্দ্র কিশোর রায় এবং মায়ের নাম আরতি রায়। ১৯৬৯ সনে তিনি প্রথম "বালক গদাধর" নামে একটি চলচ্চিত্রে শিশুশিল্পী হিসেবে অভিনয় করেন, যার পরিচালক ছিলেন হরিময় সেন। তিনি কলকাতার এ পি জে ,পার্ক স্ট্রীট শাখার স্কুলের ছাত্রী ছিলেন, যদিও নাচের প্রতি অত্যধিক ঝোঁক থাকার কারণে বেশিদূর এগোতে পারেন নি। তার ডাক নাম চুমকি, গণচীন সরকারের ১৯৫১ খ্রিষ্টাব্দে স্বাক্ষরিত চুক্তিকে বোঝানো হয়, যার ফলে তিব্বত গণচীনের অধীনে একটি স্বয়ংশাসিত অঞ্চলে পরিণত হয়।]	রায়পুর।	Success

5.1 Operational Efficiency and Storage Costs

For real-world deployment, specifically in Retrieval-Augmented Generation (RAG) systems where vector database costs scale linearly with dimensionality, the trade-off between storage footprint and retrieval accuracy is paramount. We analyzed this trade-off by comparing the physical storage requirements against NDCG@10 scores across varying dimensions on our test corpus of 1,124 documents.

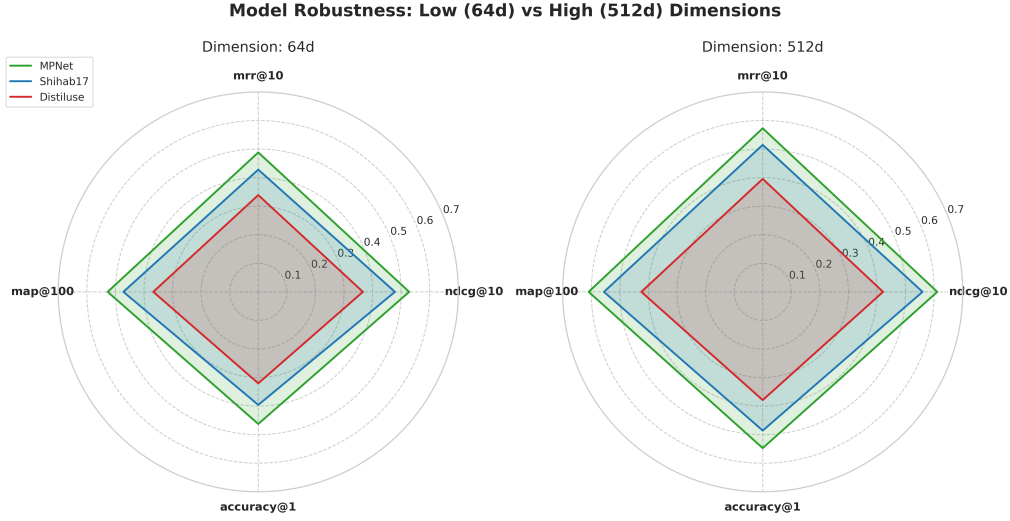


Fig. 8. Radar charts comparing model robustness at Low (64d) vs High (512d) dimensions. Even at extreme compression (64d), the MPNet model (green area) covers a larger surface area across metrics than the full-precision Distiluse model.

Figure 9 illustrates the Pareto frontier of efficiency. The fine-tuned mpnet model demonstrates remarkable resilience to quantization. As detailed in Table 5, reducing the dimensionality from 768d to 128d results in an 83.3% reduction in memory usage (dropping from 3.29 MB to 0.55 MB). Despite this massive compression, the model retains 96.1% of its peak performance (0.7796 vs 0.8114 NDCG).

Crucially, even at the extreme compression of 64d, the fine-tuned MPNet achieves an NDCG of 0.7438, which is significantly higher than the full-precision (768d) performance of the original pre-trained base model (0.4715). This confirms that our Matryoshka fine-tuning strategy not only aligns the vector space for Bengali but does so in a way that concentrates semantic information into the earlier dimensions, identifying 128d as the ideal configuration for resource-constrained production environments.

Table 5. Storage vs. Performance Analysis. Comparison of the Base model against the Fine-Tuned (FT) MPNet model across compression levels. At 128d, the model retains 96% of its performance while reducing storage by 83%.

Model Config.	Dim.	Size (MB)	NDCG@10	Retention
MPNet-Base	768d	3.29 MB	0.4715	-
MPNet-FT	768d	3.29 MB	0.8114	100.0%
<i>Compression Analysis</i>				
MPNet-FT	512d	2.20 MB	0.8072	99.5%
MPNet-FT	256d	1.10 MB	0.7918	97.6%
MPNet-FT	128d	0.55 MB	0.7796	96.1%
MPNet-FT	64d	0.27 MB	0.7438	91.7%

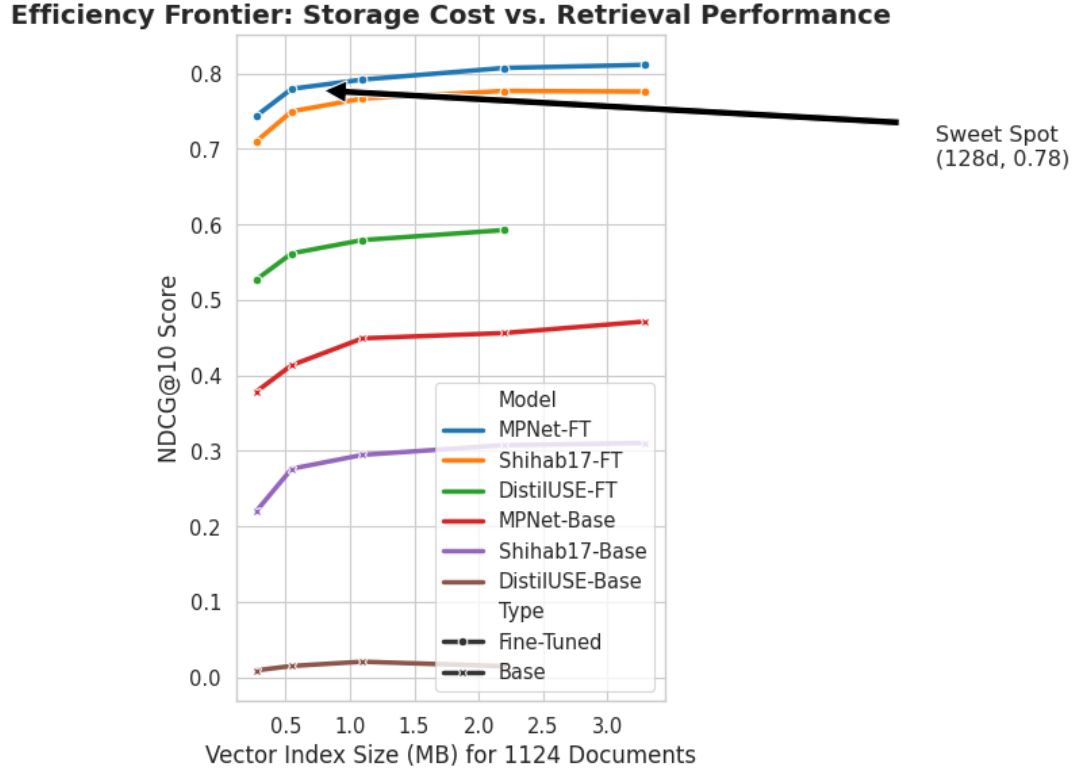


Fig. 9. **The Efficiency-Accuracy Trade-off.** The X-axis represents the memory footprint (MB) required to index the test corpus ($N = 1124$), while the Y-axis shows retrieval performance (NDCG@10). The fine-tuned mpnet-ft model (blue line) exhibits a superior performance curve. The annotation highlights the operational “sweet spot” at 128 dimensions, offering high accuracy with an 83% reduction in storage costs compared to the 768d baseline.

5.2 Response Accuracy and Hallucination

The analysis reveals that MPNet’s superior retrieval metrics translate directly to better generation. As shown in Figure 8, the MPNet profile (green) encompasses the others even at 64 dimensions.

Consider the query: *“Who was the director of the world’s first film?”*

- The shihab17 and distiluse models failed to retrieve the correct paragraph regarding the Lumière brothers. Consequently, the generation component hallucinated.
- The mpnet model correctly identified that the information was missing and responded with “I don’t know,” resulting in a high Faithfulness score.

Figure 10 further correlates model confidence with retrieval success, indicating that mpnet embeddings are more semantically precise, effectively distinguishing between topical relevance and factual answers.



Fig. 10. Heatmap illustrating the correlation between retrieval confidence and generation accuracy across different query types.

5.3 Error Analysis and Model Limitations

While the quantitative metrics highlight mpnet-ft as the superior model, a granular analysis of failure cases reveals distinct behavioral patterns intrinsic to each architecture. We categorized 100 error samples from the test set into three primary failure modes: *Lexical Overlap Bias*, *Entity Hallucination*, and *Tokenization Granularity*.

Lexical Overlap Bias: The distilled model (distiluse) frequently succumbed to lexical overlap bias, retrieving contexts that shared high keyword frequency with the query but lacked semantic alignment. For example, for the query "What is the currency of Bangladesh?", distiluse retrieved passages discussing "Bangladesh's economic growth" rather than the specific currency entity. In contrast, mpnet, with its deeper interaction layers, successfully mapped the query to contexts containing the specific answer "Taka", even when the word overlap was lower.

Tokenization and Monolingual Advantage: Although mpnet dominated overall, the monolingual shihab17 model exhibited superior performance on queries involving rare Bengali idioms or complex agglutinative morphology. Table 6 illustrates a specific case where mpnet failed. The compound word 'মুক্তিযুদ্ধভিত্তিক' (Liberation War-based) was split into four sub-tokens by the multilingual tokenizer of MPNet, diluting its semantic representation. The monolingual tokenizer of shihab17 treated it as a more cohesive unit, allowing it to retrieve the correct literary context.

False Negatives in High Compression: At extreme compression (64d), mpnet occasionally failed to distinguish between negation and affirmation. In queries asking "Which feature is NOT present in...", the 64-dimensional vector

lost the specific dimension encoding the negation, retrieving positive instances instead. This suggests that while Matryoshka learning is efficient, syntactic nuances like negation require higher dimensional bandwidth (minimum 128d).

Table 6. Error Analysis: Instance where Monolingual Model outperforms MPNet due to Tokenization

Query (Trans.)	মুক্তিযুদ্ধভিত্তিক উপন্যাসের নাম কী? <i>What is the name of the Liberation War-based novel?</i>
Correct Context	...আনোয়ার পাশা রচিত 'রাইফেল রোট আওরাত' একটি উল্লেখযোগ্য মুক্তিযুদ্ধভিত্তিক উপন্যাস।...
MPNet-FT (Reason)	Failure: Retrieved a passage about general Bengali literature history. Tokenizer fragmentation of 'মুক্তিযুদ্ধভিত্তিক' resulted in loss of specific "war" context.
Shihab17-FT (Reason)	Success: Correctly retrieved the passage regarding 'Rifle Roti Aurat'. Optimized Bengali vocabulary retained the compound word semantics.

6 Limitations

While our results are promising, this study has several limitations. First, our fine-tuning and evaluation are restricted to the *BanglaRQA* dataset, which primarily consists of factoid questions from Wikipedia. The models' performance on conversational queries, abstract reasoning, or domain-specific texts (e.g., medical or legal Bengali) remains untested.

Second, while we demonstrate the efficiency of Matryoshka Representation Learning, we did not perform an ablation study to quantify the specific contribution of the MRL objective versus the Multiple Negatives Ranking Loss alone.

Finally, our error analysis indicates that while mpnet is superior, it still struggles with negation and complex morphological compounds at high compression rates (64d), suggesting that a minimum dimensionality of 128d is a hard constraint for semantic nuance.

7 Ethics Statement

This work utilizes the publicly available BanglaRQA dataset, derived from Wikipedia, which does not contain personally identifiable information (PII) or offensive content. Our models are intended to improve information access for Bengali speakers. However, we acknowledge that retrieval models can inadvertently surface biased or hallucinated content if the underlying knowledge base is flawed. All models and code will be released open-source to promote transparency. Regarding environmental impact, our experiments were conducted on a single GPU for less than one hour, resulting in a negligible carbon footprint compared to training Large Language Models from scratch.

8 Conclusion

This study provided a comparative evaluation of three distinct embedding architectures for Bengali semantic retrieval. Our results conclusively demonstrate that while specialized fine-tuning improves all models, the underlying architecture sets the performance ceiling. The mpnet-base model, when fine-tuned using our proposed Matryoshka-MNRL strategy, outperforms both the monolingual and distilled models. This finding holds true not only on our primary retrieval task but also across a range of classification and retrieval tasks from the MTEB benchmark.

We further demonstrated that high-dimensional embeddings are often unnecessary for effective retrieval in Bengali. Our saturation and retention analysis confirms that models can maintain over 95% of their peak performance even

when compressed to 128 dimensions. For practitioners building Bengali RAG applications, this suggests that the mp-net architecture, combined with dimensionality reduction, offers the optimal balance of accuracy and computational efficiency.

Acknowledgments

The authors acknowledge the open-source contributions of the model creators for shihab17/bangla-sentence-transformer, sentence-transformers, and the curators of the BanglaRQA dataset.

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A Reproducibility Details

To ensure the reproducibility of our results, we detail the hyperparameters and training infrastructure used in this study. All experiments were conducted using the sentence-transformers library (v3.0+) and the PyTorch framework.

A.1 Computational Infrastructure

Models were fine-tuned on a single NVIDIA T4 (16GB) GPU. The training duration for the mpnet model was approximately 45 minutes for 4 epochs.

A.2 Hyperparameters

We utilized a unified hyperparameter configuration across all three models to isolate architectural differences. We employed the MatryoshkaLoss wrapper over the MultipleNegativesRankingLoss (MNRL). The specific configuration is detailed in Table 7.

Table 7. Hyperparameters for Fine-Tuning

Hyperparameter	Value
Batch Size (Per Device)	32
Gradient Accumulation Steps	4
Effective Batch Size	128
Number of Epochs	4
Learning Rate	2×10^{-5}
Scheduler	Cosine Decay
Warmup Ratio	0.1 (10%)
Optimizer	AdamW (Fused)
Max Sequence Length	512 Tokens
Matryoshka Dimensions	{768, 512, 256, 128, 64}
FP16 / BF16	Disabled (FP32)

A.3 Data Splitting

The BanglaRQA dataset was processed as follows:

- **Training Set:** Combined original Train and Validation splits, filtered for `is_answerable=1`. Total pairs: 8,573.
- **Test Set:** Original Test split, filtered for `is_answerable=1`. Total pairs: 1,405.